

Chunking Experiment

DocuChat retrieval and grounded-answer evaluation source

Experiment scope

Comparison of recursive, semantic, and Markdown chunking with two embedding models.

Chunking methods

Recursive chunking

Splits a document using structure such as sections, paragraphs, and sentences, and uses smaller context when needed.

Semantic chunking

Uses embeddings to detect changes in topic or meaning and splits according to those transitions.

Markdown chunking

Splits according to Markdown structure, including headings, subheadings, sections, tables, and lists.

Embedding models and results

The experiment applied embeddings using Qwen 3 8B and OpenAI Text-Embedding-3-Small.

Embedding model	Highest MRR and Hit@3
Qwen 3 8B	Recursive text splitter
OpenAI Text-Embedding-3-Small	Markdown text splitter

Metric definitions

Hit@3: The percentage of correct results found in the top three search results.

MRR: A metric that evaluates where the correct answer appears in the results.